

REPRODUCIBLE SYSTEMS AND EVALUATION REPORT

Certificate-Gated Dynamic Routing under Compound Urban Disruptions

A Chennai-oriented integration and evidence package

Execution status: Stage 10 partial, with explicit experiment gaps

Evidence regenerated September 2026

Abstract

This work implements and audits a Chennai-oriented routing framework that joins a reproducible dated OpenStreetMap road graph, historical flood evidence, scenario road-state overlays, an integer-metric Customizable Contraction Hierarchy (CCH), certificate-gated metric refresh, projected-load reservations, public-facility accessibility, and an emergency-priority policy. The regenerated graph contains 155,345 nodes and 331,545 directed arcs; two independent builds produced identical sorted node- and arc-ID digests. On 24 city-graph differential queries, unpacked CCH costs matched Dijkstra exactly. A historical-hotspot-conditioned blockage scenario disconnected about 9,965 of 3.59 million represented WorldPop residents (0.277%) in the accessibility projection. Synthetic four-node reservation experiments showed no improvement at 25% or 50% compliance, but lower mean BPR travel time at 75% and 100%; a paired emergency scenario reduced arrival time by 60 seconds while adding 15 seconds of ordinary-user external delay. These are integration and functional-evaluation results, not a calibrated 2015 flood reconstruction or operational Chennai traffic outcomes. Most OSM speeds and lanes are missing and replaced only by labelled scenario defaults; matched citywide SUMO scenarios and several planned ablations remain unexecuted. The contribution is therefore the reproducible integration and evidence discipline, with an established route-cost certificate used as a CCH refresh gate - not a new shortest-path algorithm.

1. Problem Statement

Road conditions in Chennai can change during monsoon flooding, incidents, and congestion. A road may remain physically connected while losing speed or capacity; redirected vehicles may then overload the remaining alternatives. Recomputing every route after every small update is expensive, but using stale road costs for too long can produce poor routes.

The research problem is:

How can a routing system update many Chennai vehicle routes under changing flood, incident, and traffic conditions while controlling computation, route error, unnecessary switching, and congestion created by the recommendations themselves?

The system uses a fixed directed road graph with edge costs that change between routing epochs. This is **snapshot-dynamic routing**, not formal time-dependent routing unless edge cost becomes a function of edge-entry time inside one query.

2. Objective

Develop and evaluate a reproducible framework that:

1. maps Chennai road, flood, rainfall, incident, and traffic evidence to explainable road states;
2. represents disruption through road availability and effective capacity;
3. converts flow and capacity into travel time using the BPR model;
4. uses Customizable Contraction Hierarchies (CCH) for repeated shortest-path queries;
5. avoids unnecessary CCH metric refreshes with a bounded-staleness certificate;
6. controls route changes using minimum-degradation, minimum-gain, and cooldown rules;
7. includes projected rerouting demand before assigning further routes;
8. evaluates critical-facility access, population-weighted loss, partial compliance, and an emergency scenario; and
9. compares all outcomes with Dijkstra and other justified baselines.

3. Expected Outcome

The expected outcome is an evaluated research prototype—not an operational navigation service. It should establish:

- whether certificate-gated synchronization reduces CCH refresh work without violating its declared route-quality bound;
- when CCH is preferable to repeated Dijkstra for the observed query/update workload;
- whether stability and projected-load controls reduce route churn and detour congestion;
- how compound disruption changes access to hospitals, fire stations, and relief centres;
- how benefits change under incomplete guidance compliance and uncertain data; and
- which conclusions are supported only by synthetic experiments versus Chennai evidence.

4. Research Gap and Contribution

4.1 Novelty Verdict

The project does **not** introduce a new shortest-path algorithm or a new approximation-certificate principle.

The following are established:

- Dijkstra, A*, ALT, CCH, CATCHUp, LPA*, and D* Lite;
- BPR travel-time functions and capacity reduction;
- flood-aware routing and flood–SUMO coupling;
- dynamic rerouting thresholds, bounded rationality, and cooldown;
- projected-load or route-reservation routing;
- emergency priority;
- lower-bound/upper-bound certificates for bounded-suboptimal paths.

The strongest defensible contribution is:

A reproducible Chennai-oriented compound-disruption framework that applies an established monotone lower/upper-bound certificate as a query-level CCH refresh gate, and is designed to experimentally separate routing-index performance from the traffic effects of stable, projected-load-aware route adoption.

Provisional novelty assessment: Medium for the proposed integration/evaluation and Low for algorithmic novelty, pending a systematic database search and completed Chennai experiments.

4.2 Closest Prior Art

Existing work	What already exists	Remaining difference
Bono et al., CPD-Search, IJCAI 2019	Old shortest distance as a lower bound, old path re-evaluated as an upper bound under increasing costs	Uses a compressed path database and A* repair, not a CCH refresh gate
Aine and Likhachev, 2016	Truncated incremental repair with bounded suboptimality	Query-specific incremental search, not batched CCH customization
Dibbelt et al., CCH, 2016	Metric-independent preprocessing, customization, fast exact queries, partial propagation	No per-query certificate for deferring refresh
Buchhold et al., 2019	CCH with BPR traffic assignment and batched queries	No flood state, stability controller, or certificate gate
Chan et al., 2023	CCH customization/query costs, recheck periods, improvement thresholds, compliance/penetration, and congestion redistribution	No flood evidence or deterministic monotone per-query stale-metric certificate
CERT-FLOW, 2026 preprint	Implemented proof-gated CH/oracle routing under drifting costs	Probabilistic conformal bounds and dual search, not the deterministic monotone specialization
Li et al., 2026	Hydrodynamic flooding, SUMO, rerouting, and emergency vehicles	No CCH certificate or explicit projected-load/stability evaluation
Pan et al., 2012	Proactive projected vehicle footprints and sequential rerouting	No flood evidence or CCH

No “first certificate,” “first CCH traffic assignment,” or “first flood-aware Chennai router” claim is supportable.

Search Protocol and Limit

The audit used targeted English-language searches through 6 September 2026 across scholarly web indexes, DOI/publisher records, surveys, reference chains, and Google Patents. Search concepts included CCH dynamic rerouting, lazy/partial customization, stale metric shortest path certificate, monotone edge-weight increase, bounded suboptimal replanning, BPR route reservation, flood SUMO routing, and Chennai flood routing. Candidate work was screened for certificate logic, routing index, traffic assignment, flood evidence, stability, and emergency/accessibility evaluation.

This was a focused prior-art audit, not a registered systematic review. The claimed gap is therefore provisional and must be rechecked using venue-specific databases and documented inclusion/exclusion counts before submission.

4.3 Chennai-Specific Gap

Chennai work already includes:

- crowdsourced flooded-street mapping ([Naik, 2016](#));
- flood-relief vehicle routing ([Ganguly and Roy, 2017](#));
- city-level Dijkstra/A*/ALT routing ([Kumar et al., 2020](#));
- flood forecasting through C-FLOWS ([publisher PDF](#));
- flood susceptibility mapping ([Alabdan et al., 2025](#));
- SUMO calibration for heterogeneous Chennai traffic ([Sashank et al., 2020](#));
- Chennai BPR-family calibration ([Gore, Arkatkar, Joshi, and Antoniou, 2023](#)); and
- a recent flood/traffic/safety navigation concept ([2026 paper](#)).

No verified Chennai study was found that jointly evaluates flood-dependent effective capacity, CCH refresh behavior, projected route load, route stability, partial compliance, and population-weighted critical-facility

access. This is an evidence-based search result, not proof of universal absence.

4.4 Publication Positioning

The paper should be positioned as an **integration, systems, and experimental evaluation paper**. The certificate is an established principle specialized to CCH synchronization. Publication strength depends on:

- transparent Chennai data provenance;
- calibrated or sensitivity-tested capacity assumptions;
- native CCH versus Dijkstra workload benchmarks;
- SUMO outcome evaluation;
- ablations isolating the certificate, CCH, stability, projected load, and priority;
- public experiment configurations and negative results.

4.5 Historical Data and Publication Validity

The absence of a public live Chennai traffic/closure feed does not by itself preclude a retrospective methods submission; suitability depends on validation quality and venue. The study must be framed as a **retrospective historical-evidence-conditioned simulation/scenario reconstruction**, not an observed road-state replay or operational live deployment.

Historical evidence can strengthen reproducibility because every method is evaluated against the same dated event. The evaluation will:

1. condition scenarios on documented Chennai flood evidence and rainfall for declared dates;
2. preserve source versions, retrieval dates, checksums, and spatial resolution;
3. separate model calibration/sensitivity from held-out scenario evaluation;
4. inject controlled 0/30/60/120-minute lags and false-positive/false-negative states into derived/simulated road-state traces;
5. use repeated SUMO seeds and report confidence intervals;
6. compare historical-evidence routing with perfect-information and no-flood baselines; and
7. describe the architecture as **designed for near-real-time operation**, not as a validated near-real-time or live Chennai service.

OpenCity hotspot/hazard layers do not provide authoritative timestamped road-state trajectories. Until such observations are obtained, lag/error tests measure robustness to controlled assumptions rather than empirical sensing accuracy. The study cannot prove present-day live accuracy or deployment readiness.

5. Proposed Method

5.1 Network and Cost Model

Let $(G=(V,E))$ be a fixed directed multigraph. Every edge has:

- free-flow time (t^0_e) ;
- baseline capacity (c^0_e) ;
- assigned entering demand $(x_{\{e,t\}})$;
- flood multiplier $(m^{\{flood\}}_{\{e,t\}})$;
- incident multiplier $(m^{\{incident\}}_{\{e,t\}})$; and
- source timestamp/confidence.

Effective capacity is:

$$c^{\text{eff}}_{e,t} = \begin{cases} 0, & \text{verified closure} \\ \max(c^{\min}_e, c^0_e m^{\text{flood}}_{e,t} m^{\text{incident}}_{e,t}), & \text{otherwise} \end{cases}$$

The BPR route cost is:

$$t_{e,t} = t^0_e \left[1 + \alpha_e \left(\frac{x_{e,t}}{c^{\text{eff}}_{e,t}} \right)^{\beta_e} \right]$$

where adjacency denotes multiplication. For software and CCH, seconds are quantized to non-negative integer milliseconds. Flow and capacity use the same interval and units. Discharged throughput is retained as an outcome, not substituted for assigned entering demand.

When $(c^{\text{eff}}_{e,t}=0)$, the edge is excluded and its routing cost is $(+\infty)$; the finite BPR expression is not evaluated. Native CCH currently rejects $(+\infty)$ closures; Dijkstra omits those edges. An integer-millisecond conversion is implemented for the Stage 2 snapshot glue.

5.2 Which Algorithm Finds the Path?

Stage 1 uses Dijkstra. Stage 2 can select Dijkstra or the native CCH adapter on synthetic graphs. Stage 6 validates the Chennai topology conversion, finite closure handling, quantization, inertial ordering, customization, unpacking, and path equality; turn restrictions remain unavailable and are explicitly excluded from the validated claim.

- **Dijkstra**: implemented correctness oracle and Stage 1 baseline.
- **CCH**: implemented and validated against Dijkstra on sampled city-graph queries, subject to the stated turn-restriction limitation.
- **ALT-guided bidirectional A***: not implemented and excluded from the executed comparison matrix.
- **Certificate gate**: decides whether CCH must be refreshed; it is not a path-finding replacement.

The repository now contains:

- RoutingKitCCHEngine: native experimental CCH adapter;
- NetworkXDijkstraEngine: exact reference adapter;
- CertifiedLazySynchronizer: certificate and refresh controller;
- EagerRefreshRouter: repeated-refresh baseline.

5.3 Certified Lazy Synchronization

Let:

- $(\bar{w})$: the edge metric currently synchronized into CCH;
- (w) : the latest authoritative/projected metric;
- (P_0) : an exact path returned under $(\bar{w})$;
- $(L=d_{\bar{w}}(s,t))$: its old optimal distance;
- $(U=C_w(P_0))$: the same keyed-edge path evaluated under current weights; and
- (ϵ) : permitted represented-cost stretch.

When every current edge weight is at least its synchronized value:

$$w_e \geq \bar{w}_e \quad \text{for all } e,$$

the old optimum is a current lower bound. The route can be returned without CCH customization when:

$$U \leq (1+\epsilon)L.$$

Otherwise, the controller customizes CCH with the complete current metric and queries again.

Three-Road Example

Assume three parallel roads:

Road	Synchronized cost
A	5.0 min
B	6.0 min
C	8.0 min

CCH selects Road A and records $(L=5.0)$.

With $(\epsilon=5\%)$:

- projected demand raises A to 5.2 min;
- allowed cost is $(1.05 \times 5.0 = 5.25)$;
- $(5.2 \leq 5.25)$, so A is certified and CCH is not refreshed.

If A rises to 5.8 min:

- $(5.8 > 5.25)$;
- the certificate fails;
- CCH is customized with $A=5.8, B=6.0, C=8.0$ and queries again.

If A rises to 7.0 min, refreshed CCH selects B at 6.0 min.

5.4 Formal Guarantee

Assumptions: fixed topology, complete atomic metric snapshots, non-negative integer represented weights, exact synchronized-engine query, and pointwise nondecreasing current weights.

For every path (P) , $(C_{\bar{w}}(P) \leq C_w(P))$. Therefore:

$$L = d_{\bar{w}}(s, t) \leq d_w(s, t).$$

Because (P_0) remains feasible:

$$d_w(s, t) \leq U = C_w(P_0).$$

If $(U \leq (1+\epsilon)L)$, then:

$$U \leq (1+\epsilon)L \leq (1+\epsilon)d_w(s, t).$$

Thus the returned path is within $(1+\epsilon)$ of the current represented optimum.

This is an application of an established upper/lower-bound certificate, not a new theorem.

5.5 Mandatory Refresh Conditions

The lower bound is invalid if any current weight drops below the synchronized weight. The implementation refreshes before querying after:

- flood or incident recovery;
- capacity increase;
- road reopening;
- any other represented cost decrease.

A closure represented as an increase is safe for the lower-bound direction, but a candidate containing the closed edge fails the upper-bound test. Native CCH currently accepts finite integer weights only; closure-sentinel behavior still requires a dedicated validation stage.

5.6 Route Adoption and Projected Load

The certificate controls metric synchronization. A separate SCENARIO filter can then decide whether a vehicle changes route:

1. current route becomes infeasible, or degradation exceeds θ_{deg} ;
2. candidate improvement exceeds θ_{gain} ;
3. cooldown has expired unless safety requires immediate action.

Items 1–3 are implemented as `decide_route_adoption`. Time-indexed reservations (items previously 4–5) remain unimplemented. The filter is not claimed to achieve traffic equilibrium.

5.7 Emergency and Public-Service Evaluation

Emergency routing remains a secondary scenario:

- blocked roads remain forbidden;
- emergency deadline/arrival time receives priority;
- delay imposed on ordinary traffic is reported;
- no signal preemption or live ambulance tracking is claimed.

Five committed evaluation studies strengthen impact without becoming route weights:

1. **Evidence freshness and uncertainty:** controlled lag and classification-error traces.
2. **Critical-facility accessibility:** travel time/disconnection to hospitals, fire stations, and relief centres.
3. **Population-weighted access loss:** distribution of access impact using WorldPop/ward weights; this is not socioeconomic equity.
4. **Partial compliance:** nearest-integer seeded cohorts target 0%, 25%, 50%, 75%, and 100%; exact percentages require a compatible fleet size.
5. **Facility-oriented criticality:** rank directed keyed arcs by newly disconnected population and unnormalised person-time added to facility access; group both directions/parallel arcs by OSM way ID before physical-road reporting.

Generic utility functions exist for portions of all five factors in `evaluation/metrics.py` and `evaluation/robustness.py`. None of the five has yet been integrated with Chennai inputs or executed as a factor-level experiment.

6. Dataset and Data Access

6.1 Factor-to-Data, Credential, and Implementation Matrix

Factor	Required data	Chennai source	Access method	API key/account	No-credential path	Implemented now	Remaining experiment work
Evidence freshness and uncertainty	A derived/simulated binary road-state truth trace plus rainfall/flood evidence	OpenCity flood records; Open-Meteo reanalysis; optional IMERG Final V07	CKAN download, HTTPS JSON, or credentialed GES DISC download	No key for OpenCity/Open-Meteo; Earthdata account authorization plus credential s/token for optional IMERG	OpenCity + Open-Meteo	Synthetic binary step-lag and false-positive/false-negative utility only	Create and justify road-state truth, timestamp alignment, multiclass/confidence mapping, then run scenarios
Critical-facility accessibility	Geolocated health, fire, and relief facilities	OpenCity health, fire, and relief; OSM POIs	Health CSV/selected KML, fire CSV/KML, relief-centre PDF, and cached Overpass response	No key or account for the listed sources	Use coordinate-bearing KML/OSM; geocode relief addresses sequentially through no-key Nominatim, then cache and manually verify	Summaries from precomputed travel-time arrays only	Acquire/cache, verify type and emergency capability, geocode relief addresses, snap facilities, compute routes
Population impact	Population count raster	Exact 2015 India 1 km R2025A STAC item	Public STAC asset/GeoTIFF download	No key or account	Use the 1 km 2015 asset to match the 2015 flood scenario	Array-level weighted mean, p90, disconnection, and threshold summaries only	Download/clip raster, verify count conservation, map cells to graph origins
Partial compliance	Vehicle IDs and declared compliance level; no observed compliance dataset	SUMO automatic-routing options	Local SUMO configuration and TraCI/libsumo	No key or account	Seeded cohort; size is $\lfloor \lfloor \text{floor} \cdot n \rfloor + 0.5 \rfloor$, so exact percentages require a compatible fleet size	Reproducible nearest-integer cohort selection only	Wire selected IDs to rerouting devices/TraCI, define non-compliant behaviour, calibrate demand, repeat seeds
Facility-oriented criticality	Chennai graph + facilities + population; no separate dataset	Derived from OSM, OpenCity facilities, and WorldPop	Local reverse multi-source shortest-path analysis	No additional key or account	Reuse the three public inputs above	Toy-tested single directed keyed-edge ranking; added cost is unnormalised person-time and disconnection is separate	Select candidates, group/close all physical-road arcs by OSM way ID, and run Chennai-scale analysis

Feasibility conclusion: a public-source acquisition path exists for all five factors without user-supplied credentials, but the complete Chennai pipeline has not yet been demonstrated. Official NASA IMERG/SRTM data cannot be acquired in a no-credential run; Open-Meteo reanalysis and non-elevation susceptibility inputs are the declared fallbacks.

6.2 API and Authentication Checklist

Source/service	Cost for research core	Key required?	Account required?	Access/authentication detail	Important restriction
OpenCity Chennai	Free public data	No	No	Dataset page and CKAN resource download	Resources vary between CSV, KML, and PDF; validate dates/schema
Public OSM Overpass	Free community service	No	No	https://overpass-api.de/api/interpreter ; use caching and a descriptive user agent	Rate/size limits; dated extracts are preferable for reproducibility
Geofabrik India OSM extract	Free download public	No	No	Download dated .osm.pbf; a separate importer is required	India-wide file is large; ODbL attribution applies
Open-Meteo Historical Weather API	Free for rate-limited non-commercial use	No	No	https://archive-api.open-meteo.com/v1/archive returns JSON	CC BY 4.0 attribution; 600 calls/min, 5,000/hour, 10,000/day, 300,000/month; no uptime guarantee
NASA GPM IMERG/GES DISC	Free	No conventional key	Yes	Authorize NASA GES DISC and use Earthdata credentials or Authorization: Bearer	Optional because the user must create/maintain the account/token
WorldPop India 2015 1 km STAC item	Free download public	No	No	Select the GeoTIFF asset from the fixed item ID; do not discover by datetime alone	R2025A is model-derived/alpha; verify version and pixel-count conservation
Eclipse SUMO/TraCI	Free open-source software	No	No	Local installation and Python/TraCI interface	Output is simulation, not observed traffic
RoutingKit CCH	Free open-source package	No	No	Install routingkit-cch Python package	Native finite-integer/turn/closure assumptions require validation

No secret, password, or bearer token is committed to the repository.

6.3 OpenStreetMap

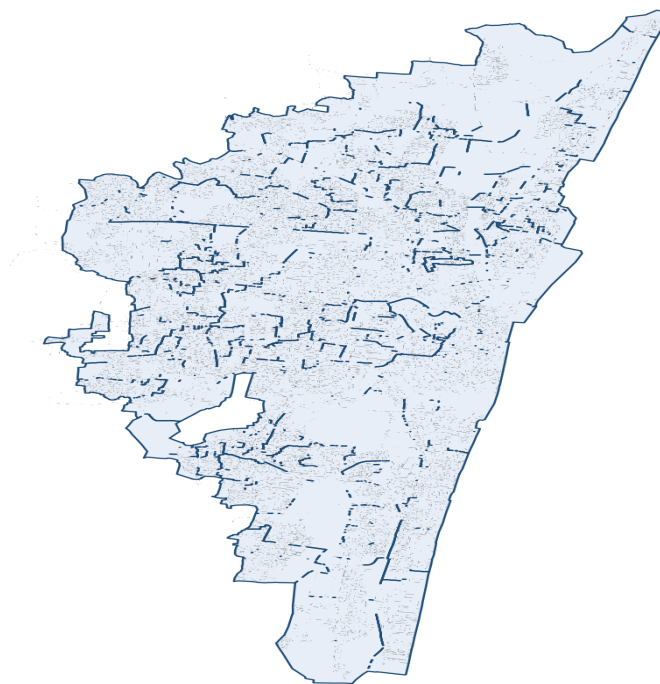


Figure 2. Stage 3 Chennai road-graph quality-assurance map.

Purpose: Directed Chennai road topology and road attributes.

Contains: Nodes, ways, geometry, road class, direction, and incomplete lanes/speeds/turn data.

Chennai coverage: Yes; completeness must be audited.

Global coverage: Yes.

Access: OSMnx/Overpass or dated Geofabrik extract.

Limitations: Mutable community data; missing attributes and turn restrictions.

Direct access: [Geofabrik India](#)

Documentation: [OSM licence](#), [Overpass API policy](#)

Repository: [OSMnx](#)

Viewer: [Chennai map](#)

The facility query will be sent to <https://overpass-api.de/api/interpreter>, cached with retrieval time/checksum, and limited to the declared Chennai study boundary. A descriptive User-Agent is mandatory. The research job will use one non-parallel request, avoid large repeated queries, and treat the public instance as best-effort with no SLA.

```
[out:json][timeout:120];
(
  nwr["amenity"~"^(hospital|clinic|fire_station)$"](12.85,80.05,13.25,80.40);
  nwr["social_facility"="shelter"](12.85,80.05,13.25,80.40);
);
out center tags;
```

The bounding box is a provisional acquisition envelope and must be replaced by the final Stage 3 study boundary. `social_facility=shelter` is exploratory OSM tagging and must not be equated with an activated GCC relief centre. The official GCC PDF remains authoritative for the declared 2024 list.

Relief-address geocoding will follow the [Nominatim usage policy](#): maximum one request/second, one thread on one machine, valid identifying User-Agent, local caching, attribution, and no recurring bulk job. Coordinates/names will be manually verified; unverified entries will be excluded.

6.4 OpenCity Chennai Flood Data

Purpose: Historical flood evidence and susceptibility validation.

Contains: The 2015 dataset has hotspots/stagnation points and a 2015 inundation zone; the separate Chennai Flooding Data collection has inundation points/depth and return-period hazards.

Chennai coverage: Direct.

Global coverage: No.

Access: Public CKAN resource download in KML.

Limitations: Historical/modelled evidence is not a current road closure.

Direct access: [Chennai Floods 2015](#), [Chennai Flooding Data](#)

Documentation: [CKAN API metadata](#)

Repository: [flood.py](#)

Viewer: Resource previews on the OpenCity page.

6.5 NASA GPM IMERG

Purpose: Historical and delayed near-current rainfall forcing.

Contains: Half-hourly satellite precipitation; historical evaluation selects Final V07 collection GPM_3IMERGHH_07.

Chennai coverage: Yes, at approximately 0.1° cells.

Global coverage: Near-global.

Access: Free Earthdata account, NASA GES DISC application authorization, and .netrc credentials or bearer-token authentication.

Limitations: Approximately 10 km cells; rainfall does not prove street flooding.

Direct access: [GPM_3IMERGHH_07 summary](#)

Documentation: [IMERG V07](#), [DOI 10.5067/GPM/IMERG/3B-HH/07](#)

Repository: `src/chennai_routing/data/rainfall.py` is currently a placeholder; acquisition is unimplemented.

Viewer: [NASA Giovanni](#)

If no Earthdata credentials are available, the no-key [Open-Meteo Historical Weather API](#) provides an ERA5-derived fallback. A [tested Chennai request](#) uses `latitude=13.0827`, `longitude=80.2707`, `declared dates`, `hourly=precipitation`, `models=era5`, and `timezone=Asia/Kolkata`. It must be labelled retrospective reanalysis—not contemporaneous sensing or road-level observation.

6.6 NASA SRTM/NASADEM and Chennai Hydrology

Purpose: Static flood-susceptibility context.

Contains: Approximately 30 m elevation plus separate drain, canal, river, and water-body vectors.

Chennai coverage: Yes.

Global coverage: Terrain is broad/global; OpenCity hydrology is Chennai-specific.

Access: Free Earthdata login and public OpenCity downloads.

Limitations: Terrain is not road-level flood depth; drain maps do not prove capacity or maintenance.

Direct access: [SRTMGL1](#)

Documentation: [OpenCity drains](#)

Repository: `src/chennai_routing/data/elevation.py` and `hydrology.py` are currently placeholders.

Viewer: [Earthdata Search](#)

6.7 Eclipse SUMO

Purpose: Dynamic traffic, queues, incidents, vehicle classes, and realized outcomes.

Contains: Software-generated vehicle/edge simulation output—not observed Chennai traffic.

Chennai coverage: User-constructed from the project graph and demand.

Global coverage: User-defined.

Access: Free open-source installation and TraCI/libsumo.

Limitations: Requires Chennai demand/behaviour calibration.

Direct access: [SUMO](#)

Documentation: [SUMO documentation](#), [automatic-routing/compliance options](#)

Repository: [Eclipse SUMO](#)

Viewer: SUMO-GUI, not a web data viewer.

6.8 Critical Facilities and Population

Purpose: Evaluate public-service accessibility and population-weighted impact.

Contains: Health-centre CSV/selected KML, fire-station CSV/KML, a 2024 relief-centre PDF, and modelled population counts.

Chennai coverage: Direct for OpenCity facilities; WorldPop covers Chennai.

Global coverage: WorldPop is global; facility catalogues are local.

Access: Public OpenCity downloads and the fixed WorldPop STAC item/GeoTIFF asset.

Limitations: UPHC/UCHC data is not a comprehensive emergency-hospital inventory; the relief PDF does not provide validated graph-ready coordinates or current activation; population exposure is not socioeconomic equity.

Direct access: [Health metadata/API](#), [fire metadata/API](#), [relief metadata/API](#), [WorldPop 2015 India 1 km item](#), [15.48 MB GeoTIFF asset](#)

Documentation: [WorldPop 1 km dataset DOI](#), [WorldPop STAC](#)

Repository: `src/chennai_routing/evaluation/metrics.py` provides array/graph utilities; ingestion and graph snapping are unimplemented.

Viewer: OpenCity resource previews and [WorldPop STAC Browser](#).

Selected OpenCity resources were downloaded and verified on 6 September 2026:

Selected resource	CKAN resource ID	Format/size	SHA-256
UPHC map	5a81427f-0aa1-4270-8df4-5d4cef250912	KML, 100,497 B	a700d7e837a5466d38e99a6ed1b67a2ff27be9638605f77b5051fef3b85055eb
UCHC map	cd4effee-997f-4fe0-b376-8801080c6962	KML, 14,155 B	45e31b7dc026f54dd4e03191c2b91278a9722af5ace5425eb44de5c1c637ed

Selected resource	CKAN resource ID	Format/size	SHA-256
Fire-station locations	39d3601b-cd42-4c11-a6c3-8fc4b057b38a	KML, 13,664 B	f41a5afb44e316aea26403f8403b08e14fcac9f13c2d5947d3e5ba859736dad9
GCC 2024 relief-centre list	ee9f087a-dc6c-41fa-8810-954a9f5887ac	PDF, 247,507 B	9d68bb27098d81b161badb721a12d7cd36ba5e856a6a033590876b61cf16ac57

These resources identify primary/community health centres, fire stations, and a relief-centre list; they do not establish trauma capability, facility capacity, opening status, or emergency readiness.

7. Data Classification and Temporal Meaning

Source	Classification	Historical/current meaning
OSM	Primary input	Mutable map snapshot
OpenCity flood	Historical evidence	Past observation/modelled hazard
SRTM/NASADEM	Supporting input	Static 2000-era terrain
Open-Meteo ERA5	No-key core rainfall input	Retrospective reanalysis, not contemporaneous sensing
IMERG Final	Optional credentialed calibration input	Historical gauge-adjusted satellite rainfall
IMERG Early	Optional credentialed near-current input	Delayed satellite rainfall estimate
SUMO	Experimental generator	Simulated traffic
Facility data	Evaluation input	Catalogue snapshot
WorldPop	Evaluation input	Modelled population
Processed edge metric	Derived dataset	Project calculation with version/time

Optional forecasts may support demonstrations, but the historical core uses no-key Open-Meteo ERA5 reanalysis and does not depend on paid APIs. No verified public live Chennai road-speed, accident, signal, or closure API is assumed.

8. Data-to-Routing Mapping

Raw source	Derived factor	Model effect	Routing effect
OSM	Topology, free-flow time, capacity assumption	Base graph	Feasible paths/lower cost
Flood history + terrain + drains	Susceptibility	Static prior	Modifies rainfall response
Open-Meteo ERA5 or optional IMERG	Rolling rainfall	Scenario-conditioning evidence	Capacity/availability assumption
Verified closure/incident	Edge state	Capacity zero/reduction	Remove/raise edge cost
Assigned SUMO demand	PCE/time entering flow	BPR $\backslash(x/c)$	Metric update
Accepted compliant routes	Projected edge-entry load	Future BPR metric	Reduces herding
Facilities + population	Access outcomes	Evaluation only	No arbitrary route penalty

9. Architecture

flowchart TD

```
Data[Road_Flood_Rain_Incident_Data] --> State[Explained_Road_State]
State --> Capacity[Effective_Capacity]
Demand[SUMO_and_Projected_Demand] --> BPR[BPR_Integer_Metric]
Capacity --> BPR
BPR --> Monotonicity[Check_Current_Weights_ge_Synchronized]
OD[Origin_Destination_Query] --> Monotonicity
Monotonicity -->|Nondecrease| StaleQuery[Query_Synchronized_CCH]
Monotonicity -->|Any_Decrease| Customize[Customize_CCH_with_Current_Metric]
StaleQuery --> PathEval[Evaluate_Old_Path_on_Current_Metric]
BPR --> PathEval
PathEval --> Gate[LB_UB_Certificate_Gate]
Gate -->|Pass| Candidate[Certified_Candidate_Route]
Gate -->|Certificate_Fail| Customize
BPR --> Customize
Customize --> FreshQuery[Query_Refreshed_CCH]
FreshQuery --> Candidate
Candidate --> Policy[Stability_Priority_Compliance]
Policy --> Reservation[Projected_Load_Reservation]
Reservation --> Demand
Policy --> SUMO[SUMO_Realized_Traffic]
SUMO --> Evaluation[Travel_Stability_Access_Evaluation]
Dijkstra[Dijkstra_Oracle] --> Evaluation
```

Implemented now: dated Chennai graph and historical-evidence ingestion; explained road states; integer BPR snapshot glue; Dijkstra/CCH engines and certificate gate; SUMO network import with synthetic demand; projected-load reservations; WorldPop/facility accessibility; and deterministic emergency policy. Matched citywide SUMO outcomes and several Stage 10 ablations remain unexecuted.

10. Implementation Status and Revised Stages

Stage 1 — Dated Historical-Hotspot Scenario Demonstration

Dated Stage 3 route response to a historical hotspot SCENARIO blockage

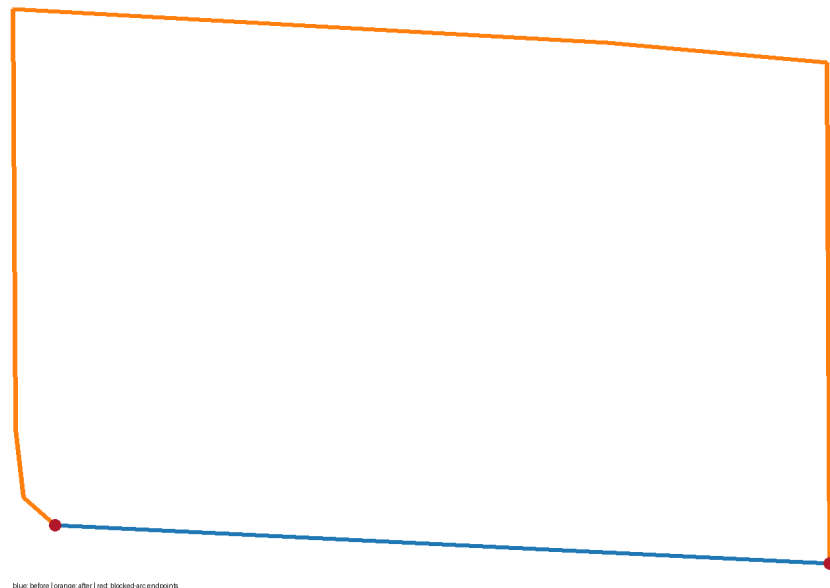


Figure 1. Dated Stage 3 route before and after the historical-hotspot scenario blockage.

Status: PASS WITH LIMITATIONS.

A documented extract of the dated Stage 3 graph, route CSV, PNG, and evidence manifest are versioned. Blocking one Stage 4 historical-hotspot arc increased the represented route cost from 9.916 to 18.687

seconds. The blockage and uniform 1200/600 capacity-flow values are SCENARIO inputs, not calibration.

Stage 2 — Certificate and Routing-Engine Validation

Status: PASS as synthetic functional evidence; rerun under Python 3.11.16.

- engine-neutral complete metric snapshots;
- exact NetworkX Dijkstra engine;
- native routingkit-cch 0.1.4 engine;
- keyed-edge path extraction;
- integer metric and atomic updates;
- bounded-staleness certificate;
- mandatory refresh after lower-bound invalidation;
- eager baseline and deterministic experiment runner;
- accessibility/compliance evaluation utilities;
- uncertainty and facility-road-criticality utilities;
- zero certificate, exact-refresh, and engine-oracle mismatches in the committed sweep.

Stage 3 — Reproducible Chennai Graph

Status: PASS WITH REPORTED ATTRIBUTE MISSINGNESS.

The pinned Geofabrik India extract passed provider MD5 verification. The GCC-clipped graph has 155,345 nodes, 331,545 arcs, one strong component, no self-loops, and no duplicate stable arc IDs. Two independent builds produced identical sorted node- and arc-ID hashes. Only 6,092 arcs have explicit speeds; 325,453 use no observed speed.

Stage 4 — Flood and Road-State Evidence

Status: PASS WITH LIMITATIONS.

OpenCity flood/drain evidence, ERA5 rainfall, and a public coarse DEM are preserved with provenance. The generated NORMAL/DEGRADED/SEVERE/BLOCKED table is a historical-evidence-conditioned scenario, not timestamped road truth or current flooding.

Stage 5 — Chennai Traffic and SUMO Calibration

Status: PASS WITH LIMITATIONS for network import; traffic outcomes remain unexecuted.

Portable Eclipse SUMO 1.27.1 imported all 331,545 Stage 3 arcs through plain node/edge files and generated 60 seeded synthetic trips. Demand, lane defaults, and most speeds are SCENARIO; no Chennai counts or OD matrix were acquired.

Stage 6 — Chennai CCH Integration

Status: PASS WITH LIMITATIONS.

Inertial CCH ordering, exact OSM-to-CCH maps, millisecond quantization, finite closure bound, full/partial customization, and path unpacking are implemented. All 24 city-graph differential queries matched Dijkstra. Turn restrictions are not modelled.

Stage 7 — Stable Projected-Load Rerouting

Status: PASS WITH LIMITATIONS on a four-node SCENARIO network.

Time-binned compliant-only reservations and update-before-certificate ordering are implemented. The matched SUMO periodic comparator is unexecuted; no equilibrium or fleet-optimum claim is made.

Stage 8 — Accessibility, Population, Compliance, and Road Criticality

Status: PASS WITH LIMITATIONS.

Verified UPHC, UCHC, and fire-station catalogues were snapped to a streaming projection of the Stage 3 graph and weighted by 2015 WorldPop. Baseline and historical-hotspot scenario accessibility, connected/all-origin p50/p90 statistics, seeded compliance, directed-arc dependency, and OSM-way grouping are reported. Relief centres were excluded because their coordinates were not independently verified; health centres are not labelled trauma hospitals and population is not equity.

Stage 9 — Emergency Scenario

Status: PASS WITH LIMITATIONS on a paired SCENARIO candidate set.

Safety/feasibility precedes deadline and bounded ordinary-user delay, with deterministic ties. Priority changed arrival by -60 seconds and ordinary-user external delay by +15 seconds in three deterministic seeds. There is no signal pre-emption, live AVL, or operational calibration.

Stage 10 — Full Evaluation and Paper

Status: PARTIAL - publication package with explicit experiment gaps.

The evidence synthesis reports executed and unexecuted baselines/scenarios/ablations, paired-seed intervals, and negative results. Citywide SUMO scenario outcomes, a matched stability ablation, and a matched population-weighting ablation remain unexecuted and are not presented as results.

11. Preliminary Certificate Experiment

11.1 Setup

- seeded synthetic directed multigraph;
- 200 nodes plus 600 additional arcs;
- 100 update epochs;
- 5 edge updates and 50 OD queries per epoch;
- 5% certificate tolerance;
- identical trace for Dijkstra and CCH;
- monotone-increase and mixed increase/decrease workloads (a changed edge decreases with probability 0.3);
- Python 3.11.16, NetworkX 3.6.1, routingkit-cch 0.1.4.

11.2 Main Results

Prototype adapter/workload	Queries	Certified stale	Lazy refreshes	Eager update refreshes	Avoided	Bound violations	Lazy total	Eager total
Dijkstra/increases	5,000	4,840	6	100	94	0	4,172.1 ms	4,701.6 ms
Dijkstra/mixed	5,000	700	86	100	14	0	2,185.5 ms	2,178.4 ms
CCH/increases	5,000	4,840	6	100	94	0	266.0 ms	429.2 ms
CCH/mixed	5,000	700	86	100	14	0	448.2 ms	432.5 ms

There were zero certificate violations, zero exact post-refresh mismatches, and zero CCH-versus-independent-Dijkstra oracle mismatches. The monotone workload produced the largest reduction because the lower bound remained valid. Mixed decreases correctly forced frequent refreshes; in this single run, lazy CCH was slightly slower than eager CCH.

These are one-machine, single-run synthetic prototype-adapter measurements with degree ordering. “Total” uses symmetric wall-clock boundaries for all post-initialization updates and route requests. Engine construction is reported separately; synchronization diagnostics combine initial and later synchronizations and should not be added to “Total.”

The CCH adapter constructs a fresh CCHMetric on synchronization and a fresh CCHQuery for each request; metric reset, partial/parallel customization, and query reuse are not benchmarked. The eager Dijkstra adapter similarly rebuilds its represented weighted graph. Therefore, these values validate prototype behavior only and do not establish an optimized CCH-versus-Dijkstra break-even point or Chennai performance.

11.3 Epsilon Sensitivity

For 2,500 CCH queries over 50 monotone epochs, refreshes fell from 33 at 0% tolerance to 25 at 1%, 4 at 5%, and 0 at 10%, with zero observed bound violations. Under mixed updates, decreases dominated: refreshes were 47, 44, 43, and 43 respectively.

Machine-readable results are stored in [docs/evidence/CERTIFIED_LAZY_SYNC_RESULTS.json](#).

12. Evaluation Plan

Baselines

1. route-once Dijkstra;
2. eager repeated Dijkstra;
3. eager CCH;
4. certificate-gated CCH;
5. ALT-guided bidirectional A*;
6. SUMO periodic rerouting;
7. independent versus projected-load-aware assignments.

Ablations

- no flood evidence;
- no certificate;

- no threshold/cooldown;
- no projected reservations;
- no emergency priority;
- no population weighting;
- compliance levels 0–100%.

Metrics

Metric	Purpose
Certificate violation and optimality gap	Method correctness
Preprocess/customize/query/unpack/epoch latency	Routing feasibility
Travel time, person-delay, queue, spillback	Traffic outcomes
Reroute count and reversals	Stability
Maximum $\lambda(x/c)$ and load concentration	Herding
Blocked/severe-edge exposure	Safety
Facility travel-time p50/p90 and disconnection	Public-service access
Population-weighted access loss	Distributional impact
Emergency deadline success and normal-user delay	Priority trade-off
Lag/error sensitivity	Data uncertainty

13. Reproducibility

```
python -m pytest
python scripts/run_stage3_boundary.py
python scripts/run_stage3_graph.py
python scripts/check_stage3_reproducibility.py record
python scripts/run_stage3_graph.py
python scripts/check_stage3_reproducibility.py verify
python scripts/run_stage4_road_state.py
python scripts/run_stage5_sumo.py
python scripts/run_stage6_cch.py
python scripts/run_stage7_reservations.py
python scripts/run_stage8_accessibility.py
python scripts/run_stage9_emergency.py
python scripts/run_certified_lazy_sweep.py
python scripts/run_stage10_evaluation.py
```

Stage 1 publication artifacts are regenerated with `python scripts/run_stage1_publication.py`. Raw and processed binaries are ignored; committed JSON manifests record provider URLs, checksums, retrieval times, software versions, classifications, and claim limits. The dated OSM PBF is pinned by provider MD5. Synthetic logical results are deterministic apart from measured timings.

14. Feasibility and Remaining Uncertainty

Public/no-key inputs and tools: OSM, OpenCity, Open-Meteo reanalysis, facility catalogues, WorldPop, SUMO, NetworkX, native CCH binding, and the tested Python utilities. Credentialed optional inputs: official IMERG and SRTM/NASADEM through Earthdata.

Uncertain/optional: public live Chennai speeds, machine-readable closures/incidents, signal-controller data, ambulance AVL, street-level satellite flood depth, and facility capacity.

The core is demonstrated as a historical-evidence-conditioned scenario reconstruction using public/no-key inputs. End-to-end topology, state mapping, SUMO network import, city CCH validation, and accessibility

have run. Calibrated Chennai demand and full citywide traffic outcomes are not demonstrated.

15. Limitations

1. The certificate principle has close prior art; algorithmic novelty is not claimed.
2. Traffic, reservation, and emergency experiments are synthetic and do not establish Chennai operational outcomes.
3. City CCH uses inertial ordering and passed cost differential tests, but timings are single-machine measurements.
4. Turn restrictions are not represented; closures use a proven finite geographic sentinel and quantized milliseconds.
5. Any weight decrease invalidates the stale lower bound and usually forces refresh.
6. The certificate controls represented route cost, not model accuracy.
7. BPR parameters and flood-capacity multipliers are not Chennai-calibrated.
8. BPR does not represent queue spillback; SUMO must measure it.
9. Historical flood observations are not live closures.
10. IMERG is coarse and delayed relative to streets.
11. SUMO demand is synthetic; no matched citywide scenario outcome or local calibration is reported.
12. Population exposure is not socioeconomic equity.
13. Facility catalogues do not provide capacity or guaranteed emergency capability.
14. Partial compliance is a sensitivity assumption, not observed behaviour.
15. Projected greedy reservations do not guarantee equilibrium.
16. A literature audit cannot prove universal novelty.
17. Accessibility and criticality use a least-cost directed-arc projection and modelled WorldPop; dependency ranking is not causal road importance.
18. OpenCity historical layers do not provide authoritative timestamped road-state truth.
19. Stage 1 uniform flow/capacity makes BPR almost a scale factor; congestion response is not validated.
20. The adoption and reservation policies are labelled scenarios, not observed driver behaviour or equilibrium models.
21. Relief-centre addresses were excluded because coordinates were not independently verified.
22. Stage 10 remains partial: matched stability, population-weighting, compound citywide, and SUMO-periodic comparisons are unexecuted.

16. Final Project Summary

Component	Final decision
Problem	Repeated routing under compound flood, incident, and congestion updates
Path engine	Dijkstra oracle and inertial CCH validated on the 155,345-node Chennai graph; ALT unimplemented and excluded
Certificate	Established LB/UB principle used as a CCH refresh gate
Defensible novelty	Chennai integration, workload characterization, and compound-disruption evaluation
Flood model	Historical OpenCity inventory + rainfall/terrain context -> explained scenario state; not road-state truth
Traffic model	BPR route estimate + imported SUMO network; matched SUMO outcomes unexecuted
Stability	Trigger, cooldown, and compliant-only time-indexed projected reservations implemented
Additional factors	Uncertainty, facility access, population-weighted loss, partial compliance, road criticality
Emergency	Secondary scenario with external-delay reporting
Current evidence	Dated reproducible graph, road-state overlay, city CCH differential, synthetic reservations/emergency, and population-weighted access
Required next evidence	Calibrated demand, matched citywide SUMO scenarios, stability/population ablations, and external validation

17. Research Claim Boundary

The completed implementation supports this statement:

A reproducible 155,345-node, 331,545-arc Chennai road graph was rebuilt twice with identical node- and arc-ID digests. On that represented integer metric, inertial CCH matched Dijkstra on all 24 sampled differential queries. Public historical/modelled evidence can be mapped to explicit scenario road states and population-weighted facility-access summaries, while synthetic reservation and emergency experiments expose compliance and external-delay trade-offs. The certificate is an established represented-cost CCH refresh gate, not a new shortest-path method.

It does **not** support:

- improved Chennai travel time or emergency response;
- operational real-time flood routing;
- a new shortest-path algorithm;
- a new approximation-certificate theorem;
- calibrated 2015 or current flood depths/closures;
- calibrated Chennai OD demand, queues, spillback, or citywide SUMO outcomes;
- trauma-hospital capability, relief-centre activation, or socioeconomic equity;
- equilibrium, optimal fleet assignment, or causal road-criticality claims;
- completion of every planned Stage 10 comparator and ablation.

The Stage 10 synthesis is therefore a transparent partial publication package. Its defensible novelty is Chennai-oriented integration, reproducibility, and evaluation design; remaining experiment gaps are enumerated in docs/evidence/STAGE10_EVALUATION_RESULTS.json rather than silently converted into

results.